

Ray Hsu

Machine Learning Software Engineer

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Professional Summary

A software developer with 6+ years of experience in Machine Learning development. Proficient in Python, PyTorch, and C/C++, with strong working knowledge of Linux and Docker. Experienced across the full deployment pipeline, including neural network training, model quantization, cross-platform porting, and real-world system integration.

Core Skills / Technical Stack

- **Deep Learning (Python / PyTorch):** Building AI models from scratch for depth estimation, object detection, and lane detection.
- **Model Optimization & Deployment (C/C++ / Bash / TensorRT / SNPE / ONNX):** Full deployment pipeline covering training, quantization, cross-platform porting, and system integration across edge AI accelerators.
- **Computer Vision & Image Processing (C/C++ / Python / OpenCV):** Multi-camera fusion, object/lane tracking, and ADAS functions (BSD, LDW, BEV, ACC).
- **Embedded Systems (Nvidia / Qualcomm / Hailo / MediaTek / Kneron):** Software development of the above functions on embedded hardware platforms.

Experience

Machine Learning Software Engineer — VIA Technologies, Inc. (2022–2026)

Automotive Embedded Systems • Model Optimization • System Integration

- **Full-Stack AI Development** (Python / PyTorch / C/C++ / Embedded System)
 - Owned the end-to-end pipeline from model training to embedded deployment, improving overall pipeline performance by 15% and shipping fully working features on edge devices.
- **Multi-Stream Object Detection Post-Processing** (C/C++ / OpenCV / Embedded System)
 - Improved object positioning accuracy by 35% and detection accuracy by 10%.
 - Designed multi-camera fusion and tracking algorithms for embedded multi-stream detection.
 - Extended functionality to support BSD (Blind Spot Detection).
- **Lane Detection Model Post-Processing** (C/C++ / OpenCV / Embedded System)
 - Designed post-processing algorithms that improved lane overlap rate by over 50%.
 - Designed tracking algorithms that improved accuracy by over 35%.
 - Developed extended functionalities including LDW, BEV, and ACC.
- **Research on Deep Learning Models for Depth Estimation** (Python / PyTorch / OpenCV)
 - Addressed the challenge of training models for dynamic objects, achieving a 70% improvement in dynamic object detection accuracy (Taiwan Patent No. TW202507568A).
 - Designed a training method leveraging the OpenCV fisheye model.
 - Designed a training algorithm for low-quality images, improving accuracy by over 50%.

Software Engineer — E-Lead Electronic Co Ltd (2019–2021)

AR-HUD • Model Design • Graphics Engineering

- **Spatial Alignment Algorithm Design** (C/C++)
 - Developed real-time spatial alignment algorithms—including 3D-to-2D coordinate transformation, curved windshield distortion compensation, and optimized dewarping—improving overall image overlay by 15%.
- **Eye-Tracking Model & System Integration** (Python / PyTorch / C/C++)
 - Established robust in-vehicle communication protocols between the HUD and external ADAS/NAVI systems to drive dynamic, real-time eye-tracking feedback.

Education

Master's Degree, Mechanical and Electrical Engineering (2015–2017)

National Changhua University of Education